

**PhD Analysis Examination**  
**September 2010**

Answer each question on a separate sheet of paper. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

Attempt EIGHT problems.

1. Let  $f \in L^1[0, 1]$ . Suppose that

$$\int_0^1 x^n f(x) dx = 0 \quad \text{for all } n = 0, 1, 2, \dots$$

Prove that  $f = 0$  almost everywhere. (Hint: reduce the problem to showing that if  $\int_E f(x) dx = 0$  for every Borel  $E$ , then  $f = 0$  a.e.)

2. Let  $(X, \mathcal{M}, \mu)$  be a  $\sigma$ -finite measure space, and let  $g : X \rightarrow \mathbb{C}$  be a measurable function. Prove that the operator  $M_g : f \rightarrow gf$  is bounded on  $L^p(\mu)$  ( $1 \leq p \leq \infty$ ) if and only if  $g \in L^\infty(\mu)$ , in which case  $\|M_g\| = \|g\|_\infty$ .
3. a) Define *monotone class*, and state the Monotone Class Theorem.  
b) State the Fubini-Tonelli theorem.
4. a) Let  $\nu$  be a signed measure on  $(X, \mathcal{M})$ . Define what it means for a set to be *positive*, *negative* or *null* for  $\nu$ , and state the Hahn decomposition theorem.  
b) Prove that if  $\nu$  is a signed measure, then there exist unique positive measures  $\nu^+, \nu^-$  so that  $\nu = \nu^+ - \nu^-$  and  $\nu^+ \perp \nu^-$ .
5. Let  $(k_n)_{n=1}^\infty$  be a sequence of measurable, 1-periodic functions on  $\mathbb{R}$  with the following properties:  
a)  $k_n \geq 0$  a.e., for all  $n$ .  
b)  $\int_{-1/2}^{1/2} k_n(t) dt = 1$  for all  $n$ .  
c) For each  $\delta > 0$ ,  $k_n \rightarrow 0$  uniformly on the set  $\delta \leq |t| \leq \frac{1}{2}$ .  
Prove that if  $f \in L^1[-\frac{1}{2}, \frac{1}{2}]$ , then  $f * k_n \rightarrow f$  in  $L^1[-\frac{1}{2}, \frac{1}{2}]$ .  
(Here  $*$  denotes convolution on  $[-\frac{1}{2}, \frac{1}{2}]$ :  $(f * g)(x) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(t)g(x-t) dt$ .)